Software Quality Assurance

Short Questions with Answers (46) · Slide by slide, from the lecture slides

Top 10: learn these first (3 short questions will come)

Q#	Question	PPT · Slide
9	What is a defect?	Quality · 19
10	What is the most cost-effective way of handling a defect?	Quality · 20, 22
19	Differentiate between process quality and product quality.	Software Quality Attributes · 4
21	Differentiate between internal and external quality attributes.	Software Quality Attributes · 7
22	What is a quality metric? Give examples.	Software Quality Attributes · 8–9
28	Differentiate between correctness and reliability.	Software Quality Attributes · 24–25
38	Define Quality Assurance and Quality Control.	SQA vs SQC · 6–7
43	Define Cost of Quality.	Cost of Quality · 2
44	Name and define the categories of Cost of Quality.	Cost of Quality · 3–7
12	Differentiate between basic, expected and exciting quality.	Quality · 23–26

★★★ most likely · ★★ may come · ★ sometimes. If QA vs QC and Cost of Quality come as long questions, the short questions are more likely from PPT 1 and PPT 2.

PPT 1: Quality

Q1. What is quality? ★★ (Slide 3–4)

There is **no single definition** of quality; different organizations define it differently. Most people understand quality as the **desirable characteristics** a product or service should have.

- Quality means **fitness for use**.
- Quality is **inversely proportional to variability**: the less a product varies, the higher its quality.

Q2. Differentiate between quality of design and quality of conformance. ★★ (Slide 5)

These are the two general aspects of fitness for use.

- **Quality of design:** products are produced in different grades or levels of quality, and these variations are **intentional**. Example: different models of cell phones, laptops and cars.
- **Quality of conformance:** how well the product **conforms to the specifications** required by the design. It is influenced by the manufacturing process, training, supervision, process controls, tests and inspections.

Q3. Name the dimensions of quality. ★★ (Slide 6–9)

The eight dimensions of quality are:

- **Performance**: will the product do the intended job?
- **Reliability**: how often does it fail?
- **Durability**: how long does it last?
- **Serviceability**: how easy is it to repair?
- **Aesthetics**: what does it look like?
- **Features**: what does it do beyond the basics?
- **Perceived quality**: what is the reputation of the company?
- **Conformance to standards**: is it made exactly as designed?

Q4. Differentiate between reliability, durability and serviceability. ★ (Slide 6–7)

- **Reliability:** how **often** the product fails. A car that needs frequent repair is unreliable.
- **Durability:** how **long** the product lasts, i.e. its effective service life.
- **Serviceability:** how **easily, quickly and economically** the product can be repaired.

Q5. Define quality according to ISO/IEC 2009. ★★ (Slide 10)

According to ISO/IEC (2009), quality is:

- the degree to which a system, component or process **meets specified requirements**;
- the ability of a product or service to **meet customer or user needs and expectations**;
- the totality of characteristics that bear on its ability to satisfy **stated and implied needs**.

Q6. Name Garvin's quality perspectives. ★★ (Slide 11)

Garvin described five perspectives of quality: **Transcendental, Manufacturing, User, Product and Value-based**.

Q7. Define any two of Garvin's perspectives. ★ (Slide 12–16)

- **Transcendental:** quality can be **recognized but not defined**, "I know it when I see it." It depends on the individual. (Tom Peters)
- **Manufacturing:** quality means **conformance to the specification**. (Philip Crosby)
- **User:** quality means **fitness for use** in the user's context. (Joseph M. Juran)
- **Product:** quality is tied to the product's **inherent characteristics ("ilities")**, e.g. reliability, usability.
- **Value-based:** quality depends on **how much the customer is willing to pay**; this leads to "good enough" quality.

Q8. What are the benefits of software quality? ★★ (Slide 17–18)

- **Fewer defects** in the software, prevented early at less cost
- **Time saving**: less rework shortens the development cycle
- **Cost saving**: fewer failures after delivery
- **Increased reliability** and fewer failure and maintenance costs
- **Fulfils actual customer needs**
- **Higher user and developer satisfaction**

Q9. What is a defect? ★★★ (Slide 19)

A defect results from a **mistake** made during the development process. Defects introduce **faults** in the software. Defects can also be **missing, ambiguous or incorrect requirements**, which lead to software that does not meet stakeholder needs.

Sequence: Error → Defect → Fault → Failure

Q10. What is the most cost-effective way of handling a defect? ★★★ (Slide 20, 22)

The most cost-effective way is to **prevent** the defect. Every prevented defect removes the rework and effort needed to fix it.

The **shorter the time** between introducing and correcting a defect, the **less rework** is required. A requirements defect that costs **1 unit** to fix in the requirements phase can cost **100+ units** if found in operations, because the cost grows **exponentially** later in the life cycle.

Q11. How is software quality accomplished? ★ (Slide 21)

- Through **process improvement**
- By increasing **staff knowledge and skill**
- Through **defect prevention** techniques
- Through better **defect detection** that finds defects earlier

Q12. Differentiate between basic, expected and exciting quality. ★★ (Slide 23–26)

- **Basic quality:** quality the customer **assumes** and does not ask for. Its presence does not satisfy, but its **absence quickly dissatisfies**. Example: tyres and steering wheel in a car.
- **Expected quality:** quality the customer **explicitly requests**. Satisfaction **increases** with quality. Example: make, model and options of a car.
- **Exciting quality: innovative, unexpected** features the customer did not know they wanted. Always in the satisfaction region. Example: cup holders when first introduced.

Q13. "Today's innovations are tomorrow's expectations." Explain. ★ (Slide 26)

Exciting features become **basic** requirements over time. Example: cup holders in cars were once **exciting quality**; now customers consider them part of **basic quality**.

Q14. Define benchmarking. ★★ (Slide 27)

Benchmarking is a process used by an organization to **identify, understand, adapt and adopt outstanding practices** from others, anywhere in the world, to **improve** its processes, projects, products and services. It helps the organization "think outside the box" and ensures its goals match **best-in-class** practices.

Q15. Name the steps of the benchmarking process. ★ (Slide 28)

1. Determine what to benchmark
2. Establish benchmarking infrastructure
3. Understand and measure current practices
4. Determine the source of benchmarking information
5. Analyze best practices
6. Plan and implement improvement

Q16. What qualities should benchmarking team members have? ★ (Slide 31)

- They should be **knowledgeable**.
- They should be **involved** in the area being benchmarked.
- They must be **familiar** with benchmarking activities.

PPT 2: Software Quality Attributes

Q17. Define software quality. ★★ (Slide 2)

Software quality is conformance to:

1. explicitly stated **functional and performance requirements**,
2. explicitly documented **development standards**, and
3. **implicit characteristics** that are expected of all professionally developed software.

Q18. What are the problems with software quality? ★ (Slide 3)

- Software specifications are usually **incomplete and inconsistent**.
- There is **tension** between customer quality requirements (efficiency, reliability) and developer quality requirements (maintainability, reusability).
- Some quality requirements are **hard to specify** clearly.
- Quality management is **not just about reducing defects**.

Q19. Differentiate between process quality and product quality. ★★★ (Slide 4)

- **Process quality** is a measure of an **activity** whose end result is a product.
- **Product quality** is a measure of the **tangible product itself**: the final executable system or intermediate products such as specifications, models and plans.

The underlying assumption is that a **quality process leads to a quality product** (CMMI).

Q20. What is a quality attribute? ★★ (Slide 5–6)

A quality attribute is a **property of a process or product** that has a qualitative or quantitative value and can be **measured or observed**.

Examples: maintainability, portability, functionality, usability.

Quality attributes apply to both the **product** (delivered to the customer) and the **process** (that produces the product).

Q21. Differentiate between internal and external quality attributes. ★★★ (Slide 7)

Basis	Internal attribute	External attribute
Derived from	The product or process **description**	The relationship between the **environment** and the system
System must run?	No, the description is enough	Yes, the system must **run**
Example	Efficiency	Reliability, robustness

Internal quality **leads to** external quality.

Q22. What is a quality metric? Give examples. ★★★ (Slide 8–9)

To know how good or bad a product is, we need **measurement**. A metric is a **qualitative or quantitative measurement scale** for a quality attribute, together with a **method or technique** to measure it.

- **Internal metrics:** complexity of code logic, performance of specific functions
- **External metrics:** tests of the system's functions, reliability and performance

Q23. What are the barriers to achieving quality? ★★ (Slide 10–11)

- Misunderstanding the **importance** of quality attributes
- Inadequate **languages** to specify quality requirements
- Inadequate **modelling methods** and notations
- **Difficulty** in designing for quality attributes
- Lack of documented **design and architecture patterns**
- **Quality control as an afterthought** in most projects

Q24. What is a quality model? Name the main quality models. ★★ (Slide 12, 15)

A quality model is a **specification of the required characteristics** that a software system must exhibit.

Main models: **Boehm** (1976), **McCall** (1977), **Adrion** (1982) and **ISO 9126** (2001).

Q25. Name the three categories of the McCall model. ★★ (Slide 16–18)

Category	Factors
Product Operation	Correctness, Reliability, Efficiency, Integrity, Usability
Product Revision	Maintainability, Flexibility, Testability
Product Transition	Portability, Reusability, Interoperability

Q26. Name the six characteristics of ISO 9126. ★★ (Slide 20)

Functionality, Reliability, Usability, Efficiency, Maintainability and Portability. These are further divided into sub-characteristics.

Q27. Name the additional factors of Boehm / the Bass framework. ★ (Slide 19, 21)

- **Boehm:** validity, clarity, understandability, modifiability, modularity, generality, economy, resilience, documentation.
- **Bass et al.: system** qualities (availability, performance, security), **business** qualities (cost, time to market) and **architectural** qualities (conceptual integrity, buildability).

Q28. Differentiate between correctness and reliability. ★★★ (Slide 24–25)

Basis	Correctness	Reliability
Meaning	System behaves according to its **specification**	**Probability** that the system operates as expected over a time interval
Nature	**Absolute** property	**Relative** property
Example	A system cannot be "almost correct"	Mean time between failures of 3 weeks

Q29. Define functionality, interoperability, security and performance. ★ (Slide 23, 27–29)

- **Functionality:** ability of the system to satisfy the purpose for which it was designed.
- **Interoperability:** ability to **work with other systems**, even ones not yet known.
- **Security:** ability to enforce authorization and authentication and to **withstand attacks**.
- **Performance: responsiveness** of the system, e.g. response time or events processed per unit time.

Q30. Differentiate between availability and recoverability. ★★ (Slide 32–33)

- **Availability:** the proportion of time the system is **up and running**.
- **Recoverability:** the ability of the system to **pick up where it left off** after a shutdown or crash. It may involve running diagnostic programs before restart.

Q31. What is usability? ★★ (Slide 34–35)

Usability, also called **user friendliness** or human factors (HCI), is the degree to which users find the system **easy to use and useful**. It depends on the target audience (novices vs. experts), is **measured using scenarios**, and is usually expressed as the **amount of time needed to learn** the system.

Q32. Differentiate internal and external understandability. ★ (Slide 36)

- **Internal:** developer understanding; contributes to **maintainability**.
- **External:** user understanding; contributes to **usability**.

Q33. What is maintainability? Name its sub-qualities. ★★ (Slide 39–41)

Maintainability is how easy it is to **change** a system after its initial release.

Sub-qualities: analysability, changeability, stability, testability.

It also refers to **repairability** (work to fix a defect), **evolvability** (work to adapt to new requirements) and **portability** (work to move to a new platform).

Q34. What is portability? ★ (Slide 42)

Portability is the ability to **reuse a component in a different application or environment**. It is a special kind of modifiability. Related attributes: adaptability, installability, conformance, replaceability.

Q35. Differentiate verifiability and testability. ★ (Slide 43–44)

- **Verifiability:** how easy it is to **verify** that desired attributes are present (internally by code inspections; externally by testing).
- **Testability: ease of testing** in terms of effort, time, complexity and cost.

Q36. Name and define the process quality attributes. ★★ (Slide 47–49)

- **Productivity:** amount of product produced for a given amount of resources.
- **Timeliness:** ability to deliver the product **on time** (short time to market).
- **Visibility (Transparency):** current process steps and project status are **accessible**.

PPT 3: SQA vs SQC

Q37. What are the two techniques for quality enhancement? ★ (Slide 3)

Quality Assurance and **Quality Control**. Both ensure that the product meets the quality expected by the **customer** and defined by the **standard**.

Q38. Define Quality Assurance and Quality Control. ★★★ (Slide 6–7)

- **Quality Assurance (QA)** is **process oriented** and focuses on **defect prevention**. It is a set of activities that ensure quality in the **process** by which products are developed.
- **Quality Control (QC)** is **product oriented** and focuses on **defect identification**. It is a set of activities that ensure quality in the **product** by finding and removing defects.

Q39. Give the ISO definitions of QA and QC. ★★ (Slide 8)

- **QA:** a part of quality management focused on **providing confidence** that quality requirements will be fulfilled.
- **QC:** a part of quality management focused on **fulfilling** quality requirements.

Q40. Why is QA proactive and QC reactive? ★★ (Slide 11–12)

- **QA is proactive** because it **prevents** defects before they occur by improving the process (e.g. benchmarking).
- **QC is reactive** because it **finds and removes** defects after they occur, through testing and debugging.

Q41. Who is responsible for QA and QC? ★★ (Slide 17–18)

- **QA: everyone** in the development team: project manager, developers, requirement analysts, testers and database designers.
- **QC:** a specific **testing team** whose objective is to test the software thoroughly to find errors.

Q42. Give examples of QA and QC activities. ★ (Slide 19–20)

- **QA:** QA audit, process documentation, establishing standards, developing checklists, internal audits, selection of tools, trainings.
- **QC:** testing, walkthrough, inspection, checkpoint review, debugging.

PPT 4: Cost of Quality

Q43. Define Cost of Quality. ★★★ (Slide 2)

Cost of Quality (COQ) attaches a **monetary figure** to the cost of not producing a high-quality product or service. **Cost of quality is the cost of defects.** It represents money that would **not be spent** if the product were developed perfectly.

Q44. Name and define the categories of Cost of Quality. ★★★ (Slide 3–7)

Category	Meaning	Example
Prevention	Cost to **prevent** defects	Quality training, process improvement
Appraisal	Cost to **identify** defects	Testing, audits, V&V
Internal Failure	Fixing defects **before delivery**	Rework, regression testing
External Failure	Fixing defects **after delivery**	Warranty, returned product, complaints

Q45. Differentiate between internal and external failure cost. ★★ (Slide 6–7)

- **Internal failure cost:** cost of handling and correcting defects found **before** delivering the product. Examples: correction of errors, rebuilding software, regression testing, rework, failure analysis.
- **External failure cost:** cost of handling and correcting defects found **after** delivering the product. Examples: corrective maintenance, complaint adjustment, returned product, warranty charges.

Q46. What is not part of COQ? How can failure costs be reduced? ★★ (Slide 8)

- Requirements, deployment, coding, shipping and installation are **not** part of COQ.
- To minimize internal and external failure costs, spend **more time on prevention and appraisal**.

How to write a short answer

- Start with a clear **definition** in one or two lines.
- If the question says "differentiate", draw a **table**.
- Add **one example** to every answer.
- **Underline** the keywords (process, product, prevent, defect, cost).